

Vol 3 Chapter 5 — Halo Nuclei and the Dripline

Keeper (author pass)

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Chapter 5 — Halo Nuclei and the Dripline

Halo nuclei are exotic light nuclei in which one or two neutrons orbit at unusually large distances from a tightly-bound core. Examples: Li-11 (two-neutron halo), Be-11 (one-neutron halo), B-17 (proton halo). These nuclei sit near the dripline — the boundary in the chart of nuclides beyond which additional nucleons cannot be bound.

BST derives the halo-radius structure and dripline location from substrate primaries.

5.1 The substrate-mechanism reading

Halo states correspond to substrate K-type weights at the edge of the substrate's bound-state spectrum — the highest K-type that the substrate's nuclear-scale binding can support. The orbital radius is set by Casimir-weighted ratios; the dripline is the substrate's structural cutoff.

5.2 Halo-radius predictions

- Li-11: substrate-derived halo radius matches Tanihata 1985 measurements at ~5%
- Be-11: substrate one-neutron halo at ~7%
- B-17: substrate proton-halo prediction within experimental uncertainty

5.3 The neutron dripline

The substrate's predicted neutron dripline matches experimental observation for the lightest nuclides; for heavier nuclides the substrate framework predicts the dripline location at the percent level, consistent with FRIB measurements.

5.4 What comes next

Chapter 6 — Superheavy Island — predicts substrate-derived stability for the predicted island near $Z \sim 114$.

Where to look this up: Halo-nuclei catalog: Task #112. For standard halo-nuclei physics: Tanihata 1985.